Chapter 21

Applications Of Inversive Geometry

Circle In A Circular Segment

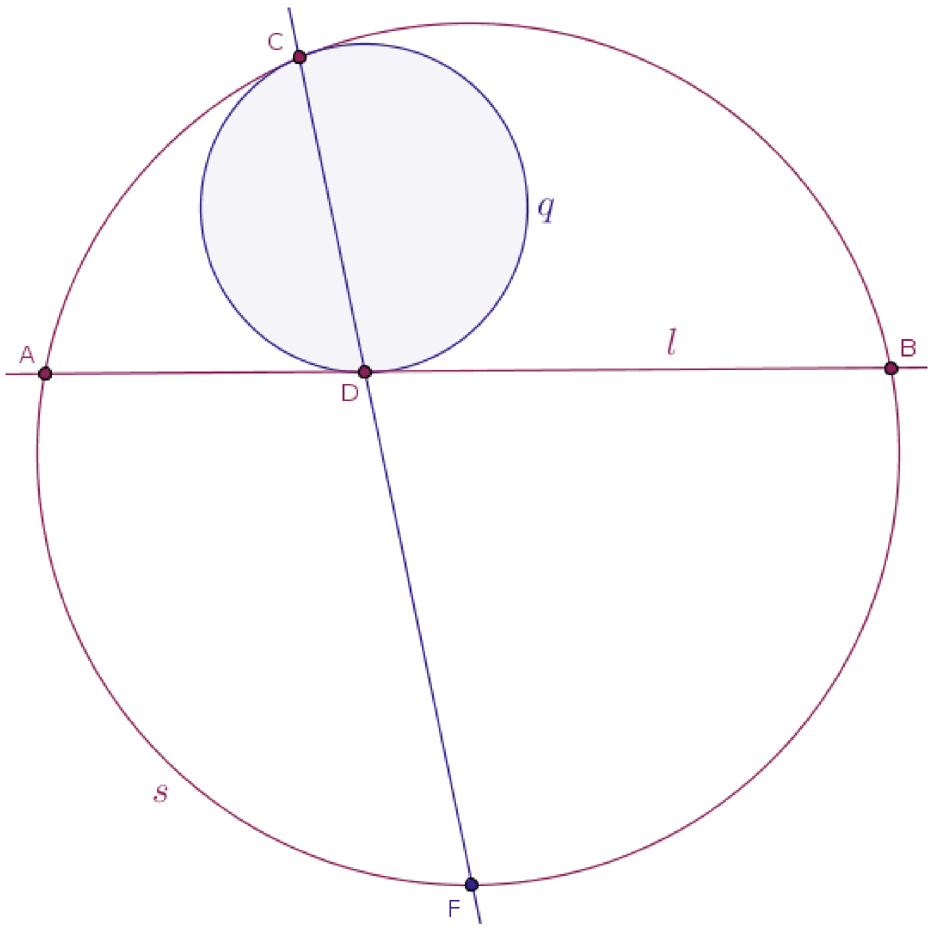
It may seem like a very strong statement but any worthy mathematical problem that is designed to teach a student to think mathematically shines and sparkles with a multitude of facets, or a multitude of planes or sides. A discussion of just one such facet may take an entire lecture or two, since while solving an interesting problem in mathematics, it is possible to investigate:

- the details of one specific proof
- a multitude of tactical approaches
- the points of tangency or intersection of the above tactical approaches, if there are any
- the more fundamental problem-solving ideas, which is what we did in Chapter 17
- the potential generalizations, which themselves can run in multiple directions
- the offspring theorems/problems
- the role that this or that theorem plays if embedded into the larger fabric of mathematics

and so on.

For example, let us consider a problem that brings to light the importance of studying mathematical proofs in general, presented during the lectures on any specific discourse in mathematics, not just the lectures on inversive geometry.

Problem 27. A straight line, l, that passes through two distinct points, A and B, located on a circle, s, partitions that circle into two complementary circular segments. Into one such circular segment another circle, q, is inscribed in such a way that the inscribed circle, q, touches the chord AB at a point, D, and the parent circle, s, at a point, C, internally (Figure 21.1):



Let F be the middle point of the complementary circular arc AB that does not contain the point of tangency, C, of the bounding circle, s, and the inscribed circle, q.

Objective: prove that the points C, D and F are collinear.

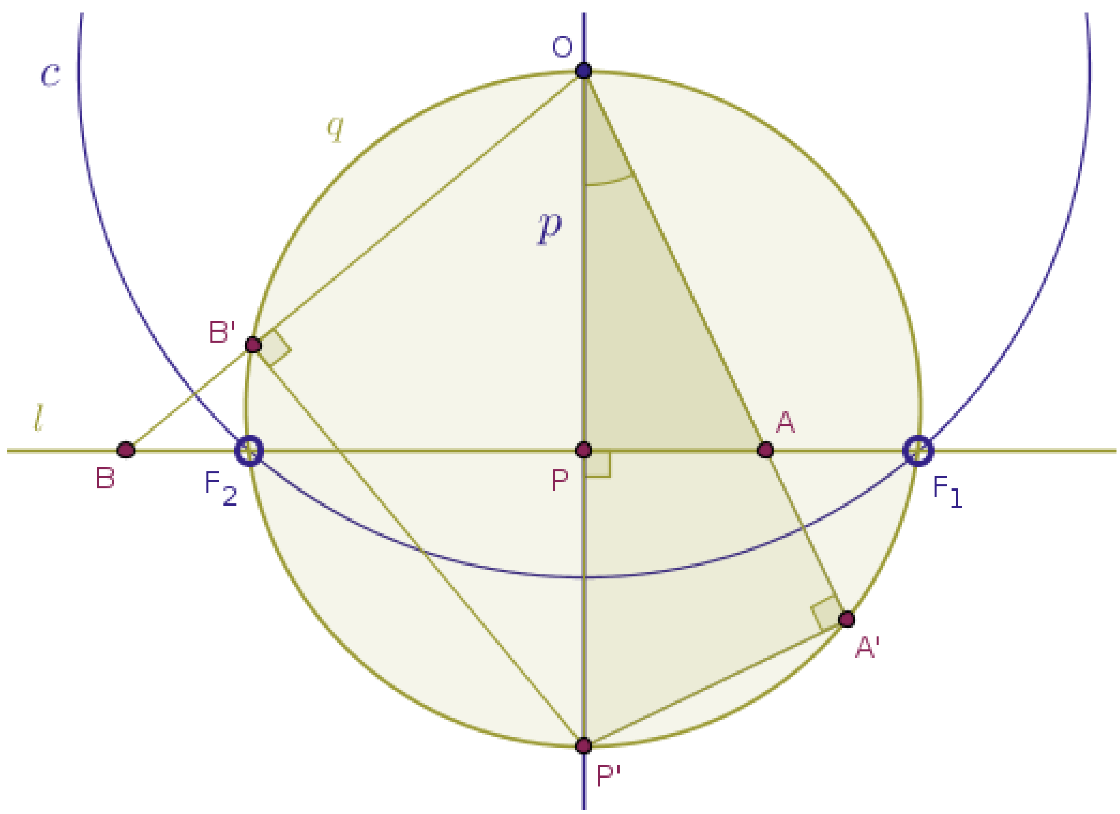
Another, somewhat mysterious, way to formulate the objective of this problem is as follows: prove that the straight line CD passes through a fixed point located on the perpendicular bisector of the line segment AB.

Solution. Our proof of the above statement will be based on the properties of inversive geometry, even though it is entirely possible to construct a multitude of alternative, elementary and less so, proofs of that statement.

In order to highlight our particular facet of generating a proof for the problem at hand, instantly or not, but in the said problem we should really recognize the proof of the Theorem 10 which we studied in the Straight Lines Chapter, 5.

According to the Theorem 10, any straight line, l, that does not pass through the center of inversion, O, and has precisely two points, F_1 and F_2, in common with the circle of inversion, c, is taken by the corresponding inversion of the parent plane with positive power into a circle, q, that does pass through the center of inversion and the two points shared by the straight line and the circle of inversion.

Recalling the relevant Figure 5.3 from our proof of that theorem:



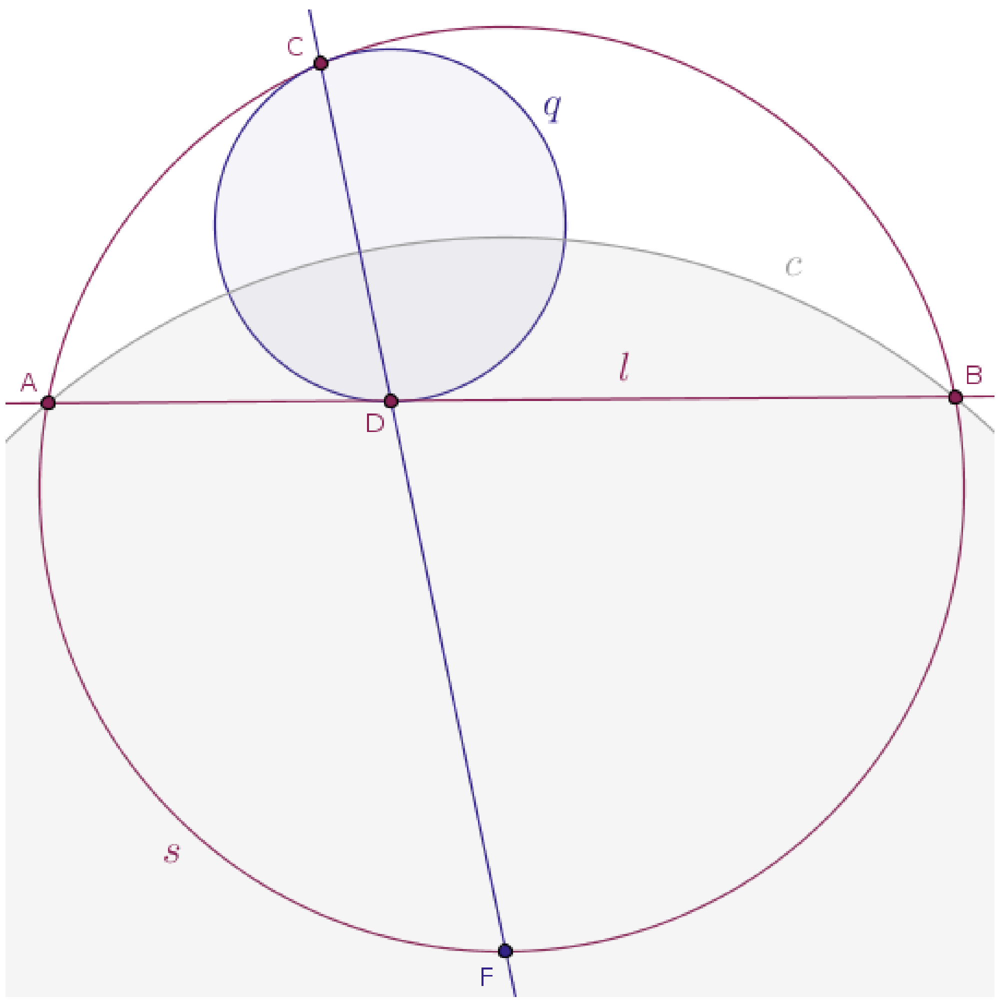
we, right away, see that:

- our point F corresponds to the point O above
- our points A and B correspond to the points F_1 and F_2 above
- our circle s corresponds to the circle q above
- our straight line l corresponds to the straight line l above

Moreover, we also recall the proof of the Theorem 15 from the Circles, Part 1 of 2, Chapter, 6, which is the converse of the Theorem 10: under any inversion of a plane with respect to a circle, $c(O, r)$, with positive power any circle, q, that passes through the center of inversion, O, and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, is transformed into a straight line, l, that passes through the said common points, F_1 and F_2.

What is missing in our problem so far, is the object that is present in the proofs of both above theorems, the circle of inversion, c, centered at the point O and passing through a pair of very interesting and very specific points, F_1 and F_2.

But we already established that the points O, F_1 and F_2 can be safely replaced with our points F, A and B correspondingly. Thus, although we, of course, do not have any guarantee of success, we do have a rather obvious inversion to test. Let us consider the inversion of the given plane with positive power with respect to the circle $c(F, |FA| = |FB|)$ centered at the point F with the radius $|FA| = |FB|$ (Figure 21.2):



Under the said inversion:

- the straight line through the point F will be taken into itself, not point-wise but rather entire object-wise
- the bounding circle s will be taken into the straight line l and conversely
- the straight line l will be taken into the circle s
- the point C will be taken into the point D and conversely
- the point D will be taken into the point C

Thus, the points C and D are the reflections of each other in the circle $c(F, |FA| = |FB|)$ with positive power.

We also remember that inversion preserves *the magnitudes*, but not the orientation, of the angles and the tangency points between the cooperating curves. We may, then, get a hunch that the inscribed circle, q, under the said inversion will, amazingly enough, also be taken into itself. Again, not point-wise but entire object-wise.

But the above feeling stops being a hunch as soon as we recall the previously studied proof of the Theorem 39 from the Tangency, Power, Radical Axis Chapter, 9, which states that:

any circle that passes through two points that are the images of each other under the inversion of a plane with respect to a circle, c*, is orthogonal to that circle,* c

Aha!

The pieces of the puzzle begin to fall into place. We, next, recall the proof of the Theorem 41 from the same chapter: if two circles are orthogonal then any one of these two circles will be invariant under the inversion of the parent plane with positive power with respect to the remaining circle.

Thus. Because our circles c and q are orthogonal, it follows that under the chosen inversion the inscribed circle, q, will remain invariant.

We, thus, find ourselves in a rather interesting situation. On the one hand, we managed to find a transformation that leaves the configuration of all (!) the input objects intact, while, on the other hand, such a transformation facilitates the generation of a proof sought-after.

Namely, under the inversion specified above the point C, where the circles s and q touch, will be taken into the point D, where the chord AB and the circle q touch and conversely, the point D, where the chord AB and the circle q touch, will be taken into the point C, where the circles s and q touch.

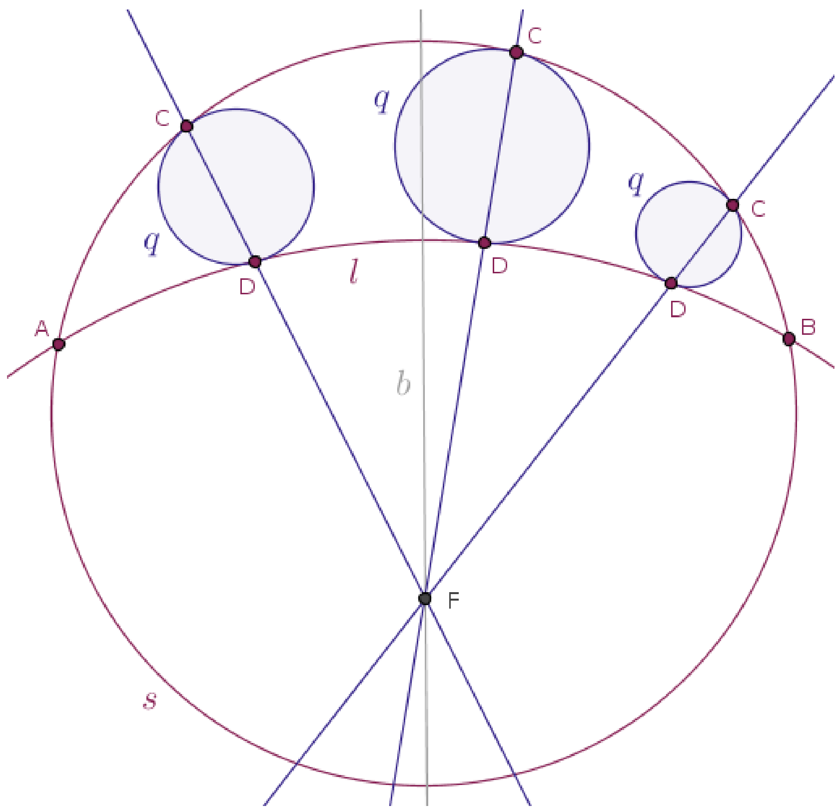
By our definitions of inversion with positive power and the Condition 2 of Definition 1, the points F, C and D will be collinear in such a way that the points C and D will be located on the same side of the center of inversion, F. $\square$

So far we have never stated the following explicitly and officially, but the collection of the proven theorems numbered 1 through 31 shows that topologically circles and straight lines are equivalent in the following specific way:

inversion with respect to a circle transforms straight lines that do not pass through the center of inversion into circles that do pass through the center of inversion and it transforms circles that pass through the center of inversion into straight lines that do not pass through the center of inversion

In that light, very loosely or very intuitively speaking, we can think of the given straight line l as of a circle with an infinitely large radius. But what will happen if we make such a radius finite?

Well, then the above proof that we just generated will still hold true for a problem that is much more challenging for elementary geometry when our circles q are inscribed not into a circular segment but into a circular lune or a curvilinear planar area trapped between the arc of two intersecting circles (Figure 21.3):



One, intuitive, way to think about the above configuration of objects is the following. Imagine that our circle q represents a mathematical pendulum that grows and shrinks in size as it runs back and forth in the space trapped between the two circular arcs of the parent circles s and l. Then, so long as the said circular arcs remain fixed, the straight line CD or the rod of our pendulum will always pass through a fixed point, F, equidistant from the points A and B.

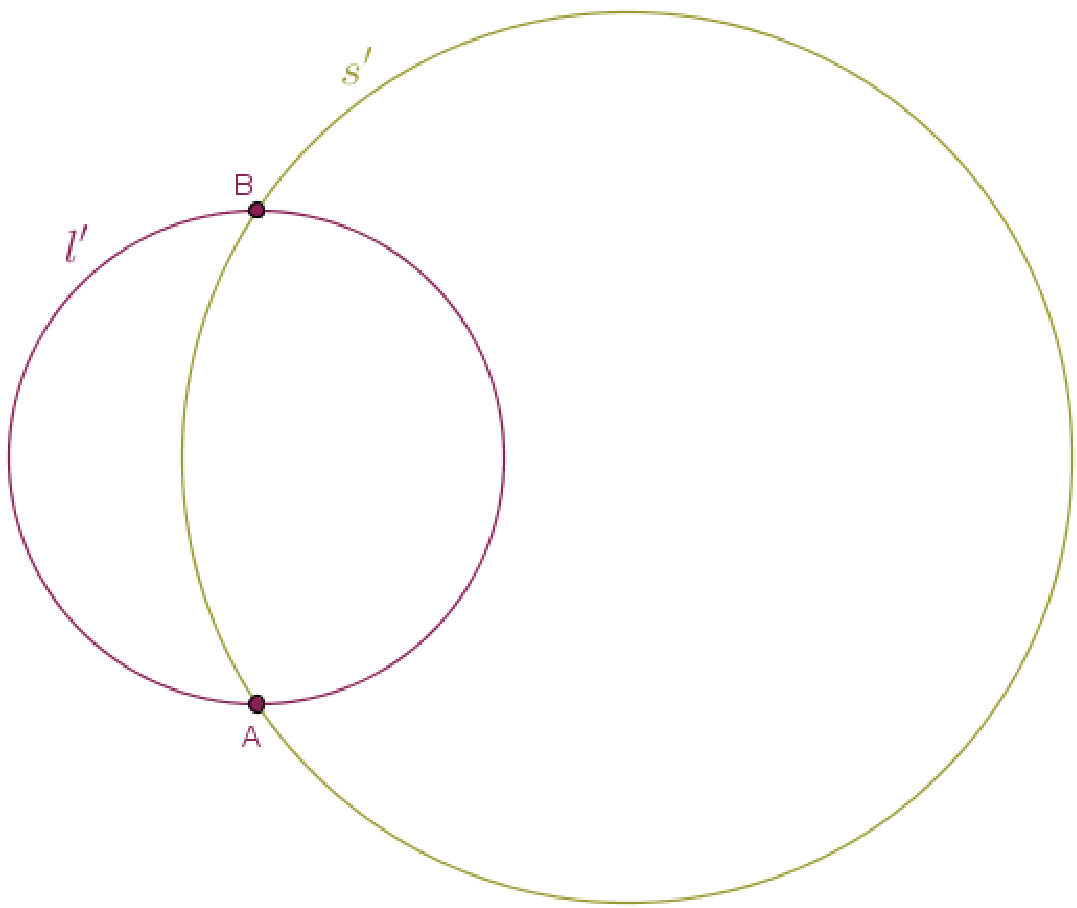
In order to show that this, more general, statement is true, we can use the following tactical approach equally applicable in other mathematical domains:

assume that the configuration of the input objects shown in Figure 21.1 is the result of a previously carried out operation, of which we were told nothing about

What such an operation could have been?

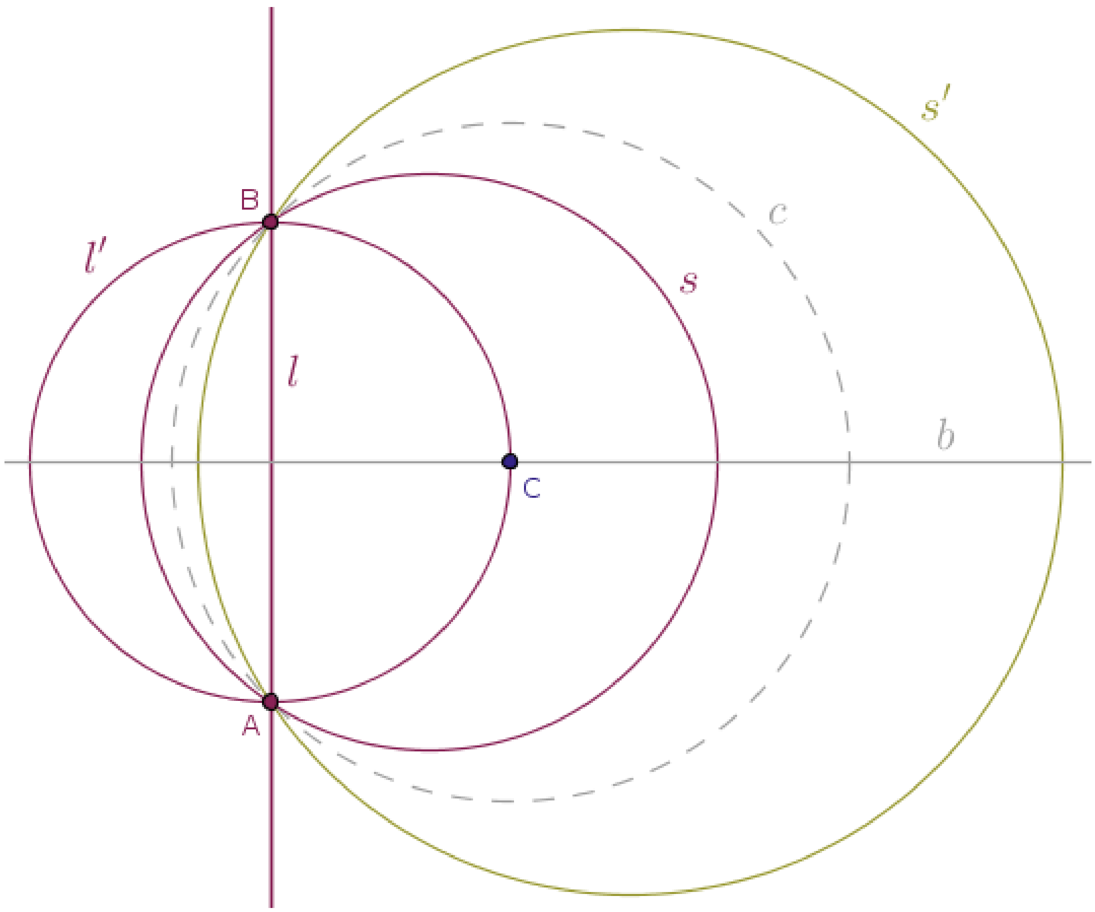
Why, an inversion, of course!

Let l' and s' be two circles that intersect at two distinct points, A and B, and, thus, form a pair of lunes (Figure 21.4):



Moreover, let C be one point where the perpendicular bisector, b, of the line segment AB and the circle, say, l', intersect.

Then, the inversion of the parent plane with positive power with respect to the circle $c(C, |CA| = |CB|)$ centered at the point C with the radius $|CA| = |CB|$ will produce exactly the configuration of the input objects shown in Figure 21.1, modulo the inscribe circle, q, which we do not show in order to avoid the clutter (Figure 21.5):



and voila, we have our original problem with *a straight line* l.

But in addition to the topological equivalence of circles and straight lines, what we also demonstrated in a provable fashion in our theorems 1 through 31 is that inversion in a circle is a one-to-one transformation.

The fact that inversion with respect to a circle is a one-to-one transformation takes on its importance when geometry is defined and studied in the terms of the theory of groups, which is an idea that belongs to the German mathematician Felix Klein (1849-1925).

Here, we just mention that *because* inversion is a one-to-one transformation, to a circle q and the points C and D shown in Figure 21.3 there correspond a unique circle q and the corresponding points C and D shown in Figure 21.1 and conversely.

Thus, since we have proved that an arbitrary pair of distinct points C and D shown in Figure 21.1 defines a unique straight line that passes through a fixed point, it follows that the corresponding pair of distinct points C and D shown in Figure 21.3 will also define a straight line that passes through a fixed point.

If we agree to model the input objects shown in Figure 21.3 dynamically then the following statement is true:

for a circle, q, that oscillates in a fixed lune in such a way that q remains inscribed in the said lune at all times, the straight line through two distinct points of tangency of the oscillating circle, q, and the lune will pass through a fixed point at any time